

Operator residual at $L = 0.3$:
off-space of A^+u_i eats the $5.6 \cdot 10^{-3}$ margin
— final 0.3 keystone checkpoint —
PRIME.RDAGGER.OPERATOR_RESIDUAL.01 — round 490

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Abstract

Last unused form of the 0.3 keystone (round 10 of the chain r478–r490). The r488 G -certificate (BS = 0.994, $N = 5$, $s = 0.05$) inverts the Galerkin R_s on V_n , not the operator R_s on $L^2[-L, L]$. The inverse-error transfer is $\|\tilde{u}_i - u_i\| \leq \|R_s u_i - \varphi_i\|_{L^2}/s$. On V_n the discrete residual is 10^{-16} ; the operator leftover is $(1 - P)A^+u_i$. **A-first.** For the load-bearing r488 solves $u_i = R_s^{-1}\varphi_i$ in V_5 , $\|(1 - P_5)A_{[t_c, 150]}^+u_1\|/s = 5.86$ against the Higham/BS margin $5.6 \cdot 10^{-3}$ (eats $\sim 1000\times$). The even ground already sits at $8.1 \cdot 10^{-2}$. Raising M to 8, 12, 20, 32 does not reduce the ratio; the $[150, 400]$ tail grows. The GL leftover of *truncated* A^+ for a full-space V_n solve is spectrally tiny at $n \geq 96$ ($3 \cdot 10^{-6}$), but that is not the residual of the certificate solutions (those live in V_5). $n = 48$ GL leftover already eats $5.5\times$. Interval Filon cannot rescue a $1000\times$ deficit. High-dps Nyström (B1) does not help: the off-space is float-visible. This form does not close. No eleventh idea. Verdict **STUCK(off/s=5.86-eats-1000x; GL-tiny-not-the-certificate;+FINAL_CHECKPOINT)**. Fourier–HS is not reused. Yoshida–Bombieri zone. No RH claim.

Status. **STUCK(off/s=5.86-eats-1000x; GL-tiny-not-the-certificate;+FINAL_CHECKPOINT)**.
No promotion of an RH claim.

Claim boundary. Nothing here is $\lambda_*(0.3) \geq c$. The operator residual of the r488 certificate solves eats the $5.6 \cdot 10^{-3}$ margin by three orders of magnitude. The 0.3 keystone is frozen. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.

1 Convention

$\text{supp } h \subset [-L, L]$, $g = h * \tilde{h}$, $\text{supp } g \subset [-2L, 2L]$, primes $n \leq e^{2L}$. $Q_W = A - P + \Pi$. At $L = 0.3$, $P \equiv 0$. A is the Fourier multiplier with symbol $\sigma_A(t) = \text{Re } \psi(\frac{1}{4} + it/2) - \log \pi$ (even, increasing on $[0, \infty)$). $\sigma_A(0) \in [-5.372183419225667, -5.372183419225665]$. t_c (unique zero of σ_A) $\in [6.289835988836902, 6.289835988836904]$. $\text{tr}(K_{t_c}) = 2Lt_c/\pi \in [1.201270186632830, 1.201270186632832]$. Even $\Pi = 2|\langle h, \cosh(x/2) \rangle|^2$, odd $\Pi = 2|\langle h, \sinh(x/2) \rangle|^2$. Fourier $\hat{h}(t) = \int h e^{itx} dx$, $\|h\|_2^2 = (1/2\pi) \int |\hat{h}|^2$, $A = (1/2\pi) \int \sigma_A |\hat{h}|^2$. Scramble-sensitive: P uses literal $\log n$; vacuous at this L . Yoshida–Bombieri zone.

2 A-first: the ratio

Write $A = A^+ - A^-$ with $A^- = |\sigma_A| 1_{|t| < t_c}$ and $R = A^+ + \Pi$, $R_s = R + sI$, $s = 0.05$. The r488 G -certificate solves $R_s u_i = \varphi_i$ in the Slepian block V_M ($M = 5$ is load-bearing; $N = 5$ GBG

$= 0.99439$, margin $5.61 \cdot 10^{-3}$). The operator residual of A^+ on those u_i is measured as

$$\text{off} = \sqrt{\|A_{[t_c, 150]}^+ u_i\|_{L^2[-L, L]}^2 - \|P_M A_{[t_c, 150]}^+ u_i\|_2^2},$$

then compared to the margin via off/s .

M	i	off/s	$\text{tail}[150, 400]/s$	$\text{IBP}[150, 800]/s$	$ u(L) $
5	0	$8.09 \cdot 10^{-2}$	$6.78 \cdot 10^{-2}$	$1.61 \cdot 10^{-1}$	0.002
5	1	5.86	7.42	$1.90 \cdot 10^2$	2.87
5	2	4.68	6.14	$1.56 \cdot 10^2$	2.35
8	1	6.10	8.42	$2.14 \cdot 10^2$	3.23
12	1	6.28	11.1	$2.24 \cdot 10^2$	3.38
20	1	6.37	16.2	$2.17 \cdot 10^2$	3.28
32	1	5.10	33.1	$3.16 \cdot 10^2$	4.78

No-Go. The certificate residual $i = 1$ is $\text{off}/s = 5.86$ versus margin $5.6 \cdot 10^{-3}$ (eats $\sim 1045\times$). The even ground already exceeds the margin. Raising M does not close the gap; the unsealed $[T, \infty)$ tail grows. $T = 400$ quadrature is anti-listed and is recorded only as a size diagnostic, not as a pad.

3 n -convergence of the GL leftover

A different residual: solve $R_s u = \varphi_1$ in the *full* Gauss–Legendre space V_n and measure the leftover of truncated A^+ against the GL interpolant.

n	GL off/s (truncated A^+)	vs margin $5.6 \cdot 10^{-3}$
48	$3.08 \cdot 10^{-2}$	eats $5.5\times$
96	$3.27 \cdot 10^{-6}$	$\sim 0\times$
128	$4.71 \cdot 10^{-7}$	$\sim 0\times$

$A^+ u$ is smooth on $[-L, L]$ ($\sigma_+ \hat{u}$ integrable; $\int \log^2(t)/t^2 < \infty$). GL projection converges spectrally: the leftover of the *truncated* operator is tiny at $n \geq 96$. That number is **not** a SATZ pad for the r488 certificate. The r488 G lives in V_5 , not V_{96} . A full-space solve loads garbage high Slepians ($|u(L)| \sim 18$); the IBP tail bound then dies at $O(1)/s$. $n = 192$ was not run: it cannot change a $1000\times V_5$ deficit, and the GL leftover is already below any honest remainder at $n = 96$. No extrapolation.

4 Why B1 does not help

High-dps Nyström would refine the *discrete* G -certificate (already residual $2 \cdot 10^{-16}$). The off-space $\|(1 - P_M)A^+ u_i\|$ is $O(1)$ in float64. Extra digits do not create a certified interpolant of garbage modes, nor a better IBP of $|u(L)| \sim 18$, nor a smaller $(1 - P_5)$ leftover. B1 is the wrong tool for this hole.

5 Checkpoint STATE (r478–r490)

What is proved, load-bearing, and not to be regressed. This is the final 0.3 keystone checkpoint (r485 was the first; r486/r488/r490 used the remaining unused forms).

Round	Verdict	Sealed content
r478	PARTIAL	even Fourier Gram PD; tail-Schur shape
r479	REDUCED	$\kappa_{\text{high}} \leq 1.79 \cdot 10^{-2}$ at $t_* = 8$, $N = 5$ (midpoint; later killed)
r480	REDUCED	refined matrix Schur; crude κ_{high} killed at 0.30
r481	REDUCED	interval FT + IBP tail; $\varepsilon = \lambda_6(K_8) \leq 1.46 \cdot 10^{-8}$, $c_{\text{tail}} \geq 0.2408$
r482	STUCK	Carleson/SR; disjoint from λ_*
r483	REDUCED	Filon[0, 150] + 4-term IBP; finite even S Higham $\mu \geq 1.20 \cdot 10^{-3}$; leftover $\ C\ _F = 2.76 \cdot 10^{-2}$ vs budget $2.4 \cdot 10^{-4}$ ($n = 48$)
r484	STUCK	20× drop falsified at $n = 96$; jets $\geq \psi_{10}$ garbage at $n = 128$; QR-IBP kills S
r485	REDUCED+CHECKPOINT	A-GO (hi-frac 0.991); finite S_w interval-PD; operator M_∞ open
r486	REDUCED	enemy 3-dim BS 0.993651; Higham $Q_E \mu \geq 2.87 \cdot 10^{-3}$; operator R^{-1} open
r487	(Lean lane)	outer bridges; disjoint from λ_*
r488	REDUCED	R not coercive; $s = 0.05$ G -cert $N = 5$ BS 0.99439; operator R -gap open
r489	(Lean lane)	outer bridges 2; disjoint from λ_*
r490	STUCK+FINAL_CHECKPOINT	off/ $s = 5.86$ eats 1000×; GL-tiny is not the certificate residual

Proved machinery (keep): frequency form of A ; t_c and $\text{tr}(K_{t_c})$; $\varepsilon/c_{\text{tail}}$ lifts; Filon and IBP integral classes; finite (flat and weighted) Schur interval-PD on V_n ; enemy Slepian numbers of K_{t_c} (0.859391, 0.313778, 0.027320, $7.70 \cdot 10^{-4}$, $1.12 \cdot 10^{-5}$, ...), first three certified; shift-invariant Birman–Schwinger ($Q \geq 0$ iff BS ≤ 1); $s = 0.05$ G -certificate $N = 5$ on V_n ; Dirichlet Q hull $[1.150 \cdot 10^{-2}, 1.192 \cdot 10^{-2}]$ (r476) hit by Higham Q_5 ($\mu \geq 2.33 \cdot 10^{-3}$).

6 Open problem (Erdős form)

The remainder is a self-contained analytic question. It does not mention the Riemann hypothesis, primes, or the von Mangoldt comb ($P \equiv 0$ at this L).

Problem 1 (Operator residual of the enemy resolvent). *Bound $\|(1 - P)A^+R_s^{-1}\varphi_i\|_{L^2[-L, L]}$ for the certified enemy modes φ_i of K_{t_c} against the Higham/BS margin $5.6 \cdot 10^{-3}$.*

Data. Let $L = 3/10$ and $s = 1/20$. Let A^+ be the Fourier multiplier on $L^2[-L, L]$ with symbol $\sigma_+(t) = \max\{\sigma_A(t), 0\}$, and Π the even/odd rank-one as above. Let $R_s = A^+ + \Pi + sI$. Let $\varphi_0, \varphi_1, \varphi_2$ be the first three eigenfunctions of K_{t_c} (Slepian numbers 0.859391, 0.313778, 0.027320; certified, no garbage). Let P be the orthogonal projection onto a finite certified subspace containing those modes (the G -certificate uses V_5). The discrete G -certificate on V_n is already < 1 (BS 0.99439 at $N = 5$). The discrete residual in V_n is $2 \cdot 10^{-16}$. The measured operator leftover on the certificate solves is $\|(1 - P_5)A_{[t_c, 150]}^+u_1\|/s = 5.86$.

Ask. Produce an explicit bound $\|(1 - P)A^+R_s^{-1}\varphi_i\|_2 \leq \delta s$ with $\delta < 5.6 \cdot 10^{-3}/3$ for each certified i , or prove that no such bound exists for any finite certified P whose G -certificate remains < 1 . Either statement freezes $\lambda_*(0.3)$.

What is already known, and may be used. R is not coercive ($\lambda_{\min}(R)$ collapses with n). The shift $s = 0.05$ is mandatory; $R_s \geq sI$. $N = 3 + |\sigma_A(0)| \cdot \text{rest}/s = 1.078$ overshoots and is not a SATZ pad. Full-space GL leftover of truncated A^+ is spectrally small at $n \geq 96$, but those solves are not the certificate. Certified jets for IBP stop at the first three Slepian of K_{t_c} ; higher Nyström modes are garbage.

What is forbidden as a pad. Fourier–HS majorants; crude κ_{high} ; integer Fourier ONB; σ^2 Plancherel as $\|Au\|_{[-L, L]}$; IBP envelope as pad; exponential-sum Ci to ∞ ; $n = 48$ leftover as

SATZ pad; $20\times$ coupling drop; QR sinc-IBP of garbage jets; flat $c_{\text{tail}}I$ as M_∞ ; constant minorant $\sigma(0)K + \Pi$; K_8 garbage jets; $T = 400$ quadrature as a tail pad; $N = 3 + |\sigma_A(0)|_{\text{rest}}/s$ as SATZ; $\lambda_{\text{max}}(BG)$ as Birman–Schwinger; the GL leftover of truncated A^+ as if it were the operator residual of the V_5 certificate; unshifted R^{-1} as coercive.

Honest type. This is an *ideas* problem, not a remaining arithmetic grind. The off-space is float-accessible and $O(1)$. Filon/IBP and GL spectral convergence are measured. What is missing is a handle on $(1 - P)A^+u$ for a u that is *both* low-mode (so IBP lives) *and* complete enough that the leftover sits under $5.6 \cdot 10^{-3}$. Those two demands have been in tension since r483. No eleventh discrete form is proposed.

7 Kills

Object	Round	Status
Fourier–HS majorant	r478	not reused
Crude κ_{high}	r480	killed
Integer Fourier ONB	r481	$Q_{0n} \sim 1/n$
σ^2 as $\ Au\ _{[-L,L]}$	r481	killed
IBP envelope as pad	r481	killed
Exp-sum Ci to ∞	r483	killed
$n = 48$ leftover as SATZ pad	r484	undercount
$20\times$ drop / block through ψ_{18}	r484	falsified
High-mode jets $n = 128$	r484	garbage $\geq \psi_{10}$
QR sinc-IBP tail	r484	$\lambda_{\text{min}}(S) < 0$
Flat $c_{\text{tail}}I$ as M_∞	r485	$d_{\text{min}}(D) < c_{\text{tail}}$
Constant minorant $\sigma(0)K + \Pi$	r479	indefinite
$T = 400$ quadrature	r488	anti-listed
$\lambda_{\text{max}}(BG)$ as BS	r488	reads 1.023
$N = 3 + \sigma_0 _{\text{rest}}/s$	r488	1.078; not a SATZ
Unshifted R^{-1}	r488	not coercive
GL-tiny at $n = 96$ as operator tail	r490	not the V_5 residual
$(1 - P_M)A^+u_i/s$ as SATZ pad	r490	5.86 eats $1000\times$

8 Scaling at $L = 0.8$

The A -measurement, Filon/IBP, finite Schur, enemy 3-dim BS, and shifted G -certificate survive (P is a cosine sum; scramble becomes live). The operator gap does not become easier: it is a completeness statement about $(1 - P)A^+u$ on a certified low-mode u , not about T or about L . The enemy-dimension observation of r486 survives: only three Slepian numbers of K_{t_c} are certified; the fourth is $7.70 \cdot 10^{-4}$. At $L = 0.8$ the same three-mode enemy is the only honest finite rank; one must additionally control P on $n \leq e^{2L}$.

9 What is not claimed

No $\lambda_*(0.3) \geq c$. The r488 G -certificate remains < 1 on V_n ; the operator residual of those solves is $O(1)$ and eats the margin. The 0.3 keystone is frozen at this final checkpoint. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.